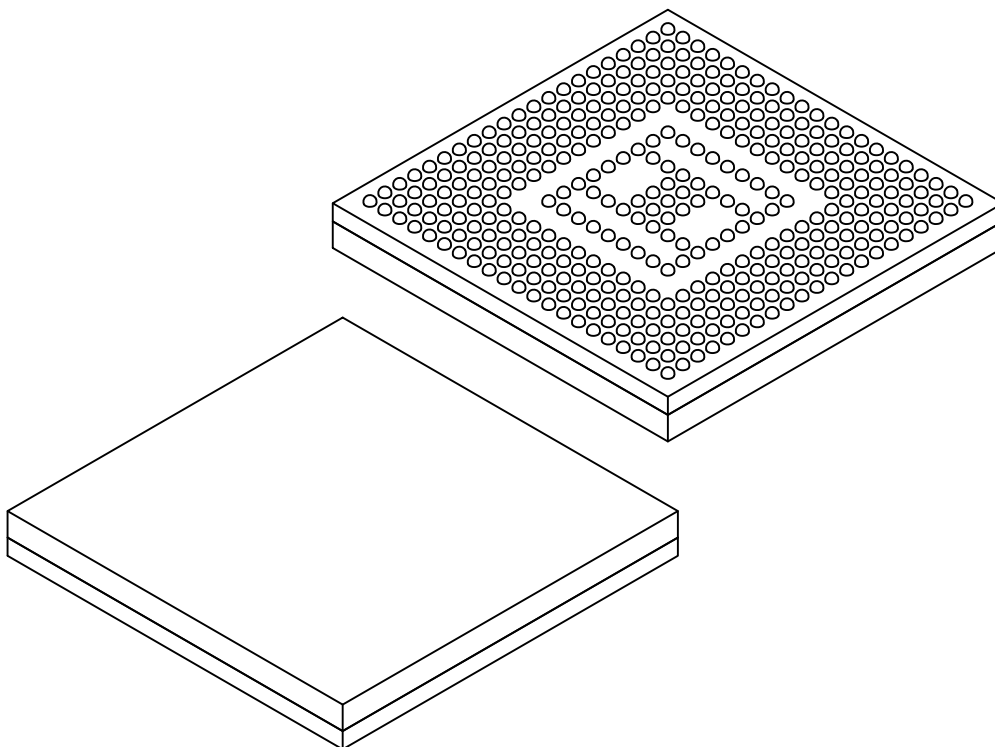


Packaging Diagrams and Parameters

375-Ball Thin Fine-Pitch Ball Grid Array Package (7EW) - 9x9x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	375		
Pitch	e	0.40 BSC		
Overall Height	A	—	—	1.20
Ball Height	A1	0.11	0.16	0.21
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.35 REF		
Overall Length	D	9.00 BSC		
Ball Array Length	D2	8.00 BSC		
Overall Width	E	9.00 BSC		
Ball Array Width	E2	8.00 BSC		
Ball Diameter	b	0.20	0.25	0.30

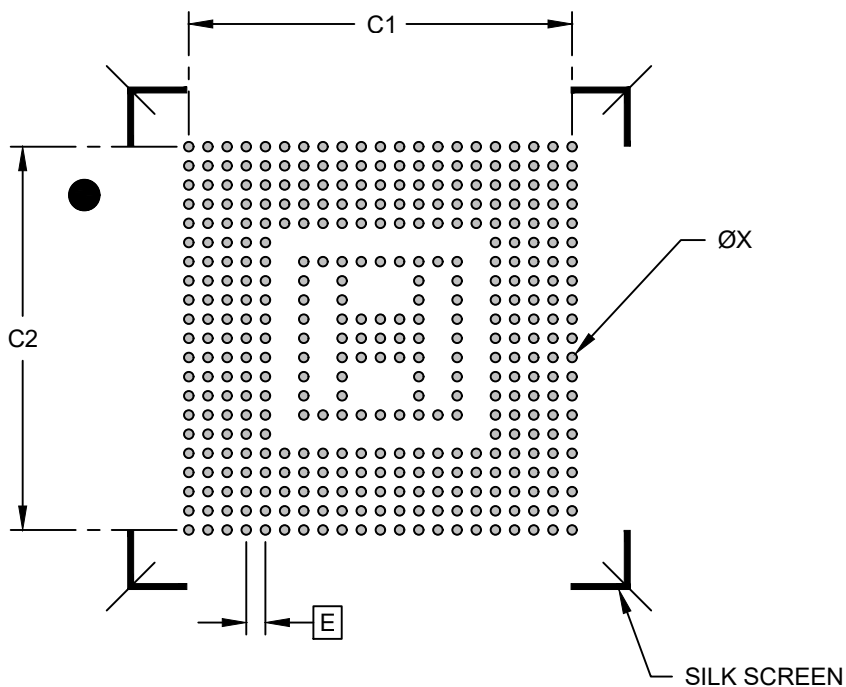
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

375-Ball Thin Fine-Pitch Ball Grid Array Package (7EW) - 9x9x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Contact Pad Spacing	C1		8.00	
Contact Pad Spacing	C2		8.00	
Contact Pad Diameter (X375)	X1			0.200

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-02644 Rev A